

Ablation Study Results

Generated: 2026-02-10T14:11:30.387192 N=150 per configuration

Configuration	Ann Acc	RCA Acc	Overall	BERT- F1	Cos Sim	Term Ov.	Avg Latency	Delta vs Full
Full System (Hybrid dual-agent baseline)	89.0%	98.0%	92.0%	0.459	0.308	0.433	5917ms	-
Single Agent - Qwen3-4B for both annotation and RCA	89.0%	94.0%	90.7%	0.457	0.309	0.401	6033ms	-1.3%
Single Agent - Qwen3-14B for both annotation and RCA	89.0%	88.0%	88.7%	0.458	0.306	0.427	5988ms	-3.3%
No Structured Output - raw text, no JSON metadata extraction	89.0%	96.0%	91.3%	0.457	0.302	0.383	6157ms	-0.7%
No System Prompt - empty system message, raw user query only	45.0%	92.0%	60.7%	0.389	0.288	0.780	11018ms	-31.3%
With Graph Context - historical episode context injected (RAG)	89.0%	90.0%	89.3%	0.454	0.288	0.335	6430ms	-2.7%
No Constitutional AI - skip validation (overhead measurement)	89.0%	96.0%	91.3%	0.457	0.307	0.420	6110ms	-0.7%